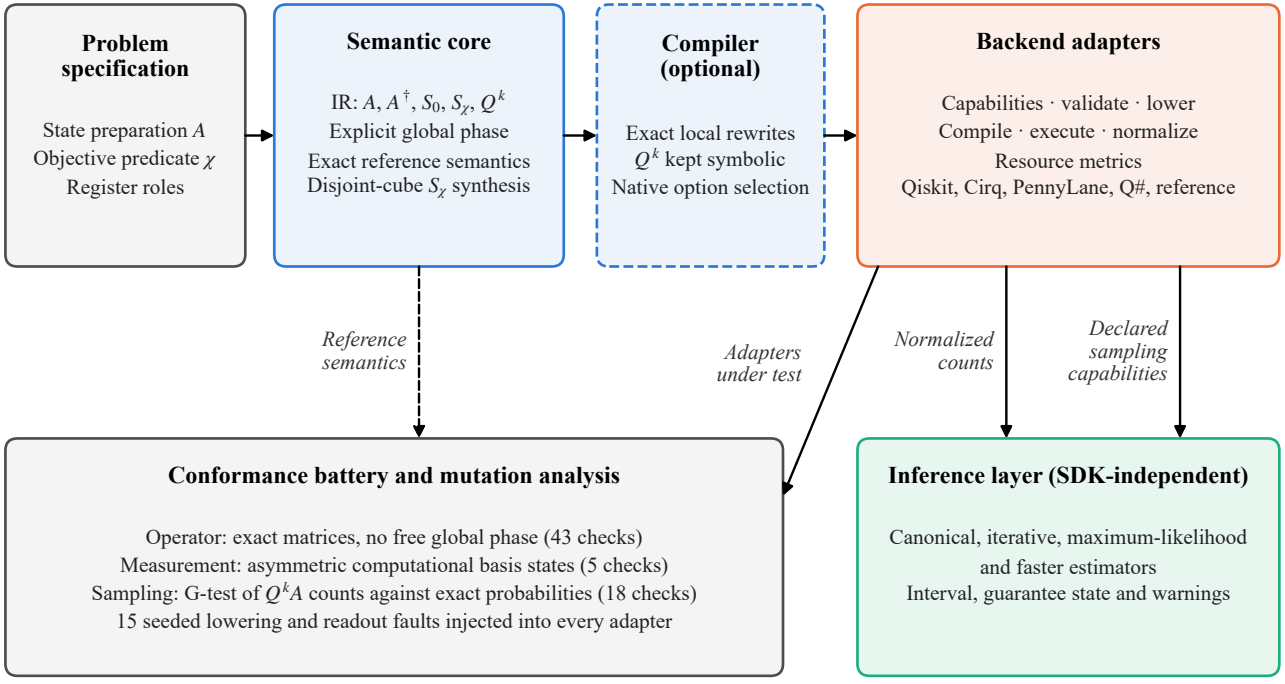

Problem specification
State preparation A
Objective predicate χ
Register roles
Semantic core
IR: A, $A^\dagger$, S_0, S_χ, Q^k
Explicit global phase
Exact reference semantics
Disjoint-cube S_χ synthesis
Compiler (optional)
Exact local rewrites
Q^k kept symbolic
Native option selection
Backend adapters
Capabilities · validate · lower
Compile · execute · normalize
Resource metrics
Qiskit, Cirq, PennyLane, Q#, reference
Reference semantics
Adapters under test
Normalized counts
Declared sampling capabilities
Conformance battery and mutation analysis
Operator: exact matrices, no free global phase (43 checks)
Measurement: asymmetric computational basis states (5 checks)
Sampling: G-test of Q^kA counts against exact probabilities (18 checks)
15 seeded lowering and readout faults injected into every adapter
Inference layer (SDK-independent)
Canonical, iterative, maximum-likelihood and faster estimators
Interval, guarantee state and warnings